

Technical Whitepaper: High-Performance Joint Optimization Solution for LORA and Sparing Based on Unconstrained Transformation

URL: <http://lora.ac.cn>

1. Industry Pain Points & Challenges

In the maintenance of high-precision equipment such as aviation and nuclear energy, Level of Repair Analysis (LORA) has long faced the "Curse of Dimensionality" and "Decision Silos":

- **Computational Inefficiency:** Traditional tools (e.g., COMPASS) suffer from exponential declines in solving speed as component counts increase in systems with multi-indenture structures.
- **Fragmented Decision-Making:** Decisions on maintenance paths ("where to repair") and spare parts configurations ("how much to stock") are typically split into two independent stages, preventing a global optimum.
- **Constraint Handling:** Real-world resource constraints (manpower, equipment, facilities) are mostly hard, non-linear, discrete constraints, causing traditional heuristic algorithms to easily fall into local optima.

2. Core Technical Solution: Constraint-to-Unconstrained Optimization Algorithm

Based on research published in the international journal *Computers & Industrial Engineering (CAIE)*, this solution proposes a new numerical solving architecture:

2.1 Constraint Softening & Smoothing Technology

Unlike traditional integer programming or branch-and-bound methods, this algorithm maps complex discrete constraints (0-1 variables) into continuous or quasi-continuous space through innovative mathematical transformations, constructing augmented Lagrangian functions or specific potential functions.

- **Advantage:** It avoids frequent constraint-conflict checks during traditional searches, enabling the algorithm to handle tens of thousands of variables.

2.2 Multi-Objective Joint Optimization Model

The algorithm integrates "Level of Repair Analysis (LORA)" and "Spare Parts Sparing" into a unified objective function.

- **Joint Solving:** Dynamically balances maintenance costs (manpower, transport) and inventory holding costs.
- **Pareto Frontier:** Instead of providing a single solution, the algorithm automatically generates an optimal trade-off curve between "Maintenance Cost vs. Delay Time".

2.3 Underpinning Acceleration via Numba

To meet industrial-grade real-time decision-making needs, the underlying code is implemented with Python + Numba for vectorized compilation acceleration.

- **Performance Benchmark:** For LORA cases of the same scale, the calculation speed is improved by [Specific Multiplier] times compared to traditional commercial software, supporting rapid reconfiguration and simulation of large-scale systems.

3. Global Strategic Value & Application Scenarios

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This belongs to "**Next-Gen ILS (Integrated Logistics Support)**" technology, addressing the needs of global high-end manufacturing (Aerospace, Defense, Nuclear Energy, Precision Instruments).

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3.1 Defense & Aerospace

- **Lockheed Martin & Boeing:** Vital for the F-35 program's "Performance-Based Logistics (PBL)" model, optimizing global supply chains and repair paths in real-time.
- **Airbus:** Reduces Aircraft on Ground (AOG) time to maximize profitability via Pareto Frontier optimization.
- **Mission Readiness:** Selects configurations with the shortest delay times during wartime through the Pareto Frontier.

3.2 High-End Manufacturing & Energy

- **ASML (Netherlands):** Optimizes global spare parts centers and engineer scheduling for lithography equipment where downtime costs are astronomical.
- **GE Vernova:** Addresses the high-cost maintenance needs of nuclear power plants and large offshore wind farms through joint optimization.

4. Conclusion & Partnership Vision

This solution has passed rigorous academic peer review (CAIE) and is ready for engineering deployment.

- **We Offer:** Core Optimization Engine API, customized LORA decision plug-ins, and algorithm consulting for complex support systems.
- **Aspiration:** To partner with top research institutes and industrial software giants to drive the generational upgrade of domestic ILS software.

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